结构化输入 Prompt

System: 你是一个脑机接口助手...
User: 解码结果+候选分布
Assistant: <待生成>

参数统计
基座模型: 4.06B (frozen)
LoRA 可训练: 9.6M (0.24%)
每层 LoRA: 266,240
36 层 × 4 targets (q/k/v/o)

Tokenizer
text → token IDs (B, L)

Token Embedding
Vocab=151,936 dim=2,560
(B, L) → (B, L, 2560)

RoPE 位置编码
θ=1,000,000

Qwen3 Decoder Layer × 36

Grouped Query Attention (GQA)

O: Linear(2560, 2560)

Scaled Dot-Product Attention + Causal Mask + RoPE

LoRA Adapters
A: (d, r) B: (r, d)
r=16, α=32, p=0.05
ΔW = (α/r) · B · A

V: Linear(2560, 640)
8 KV heads × 80 dim

K: Linear(2560, 640)
8 KV heads × 80 dim

Q: Linear(2560, 2560)
32 heads × 80 dim

RMSNorm(2560)

⊕ Residual

SwiGLU MLP

Down: Linear(9728, 2560)

SiLU(gate) ⊙ up

Up: Linear(2560, 9728)

Gate: Linear(2560, 9728)

RMSNorm(2560)

⊕ Residual

RMSNorm(2560)

LM Head
Linear(2560, 151936)
→ Softmax → next token

自回归生成
纠正后的目标文本